Hello!

Lecture 3 - Limits II

Garrett Credi

09/14/2026

Announcements

Homework Policy:

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- For homework assigned on Tueday, the due date will be Thursday @ 6PM

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- Wednesday 12:15 - 1:15 PM
- Thursday 4:30 - 5:30 PM

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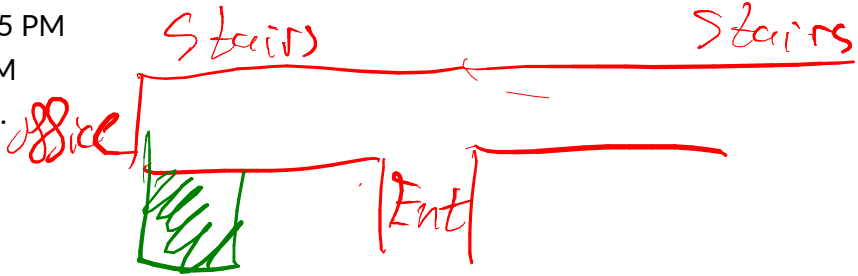
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Both are in Vincent Hall 105.



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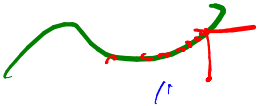
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1. Defined the notion of a *limit* of a function: "value a function approaches"

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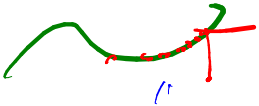
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1. Defined the notion of a *limit* of a function: $\lim_{x \to a} f(x) = L \in \mathbb{R}$ if we can make $f(x) \approx L$ up to some error ($\epsilon > 0$) just by forcing $x \approx a$ up to some error ($\delta > 0$)

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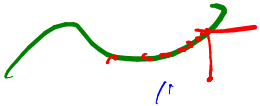
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5. Computed limits by algebraic manipulation

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Today we will develop our ability to algebraically compute limits by:

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1. Studying the "simplest" functions to take limits of, *continuous functions*
2. Discovering *limit properties* that help us reduce complicated limits to easier limits
3. Combining the above to get the process of algebraically computing limits

And we will also study cases where the limit is infinite.

"Functions whose approximate values are their true values"

Continuity

Continuous Functions I

To start off: let's consider the function:

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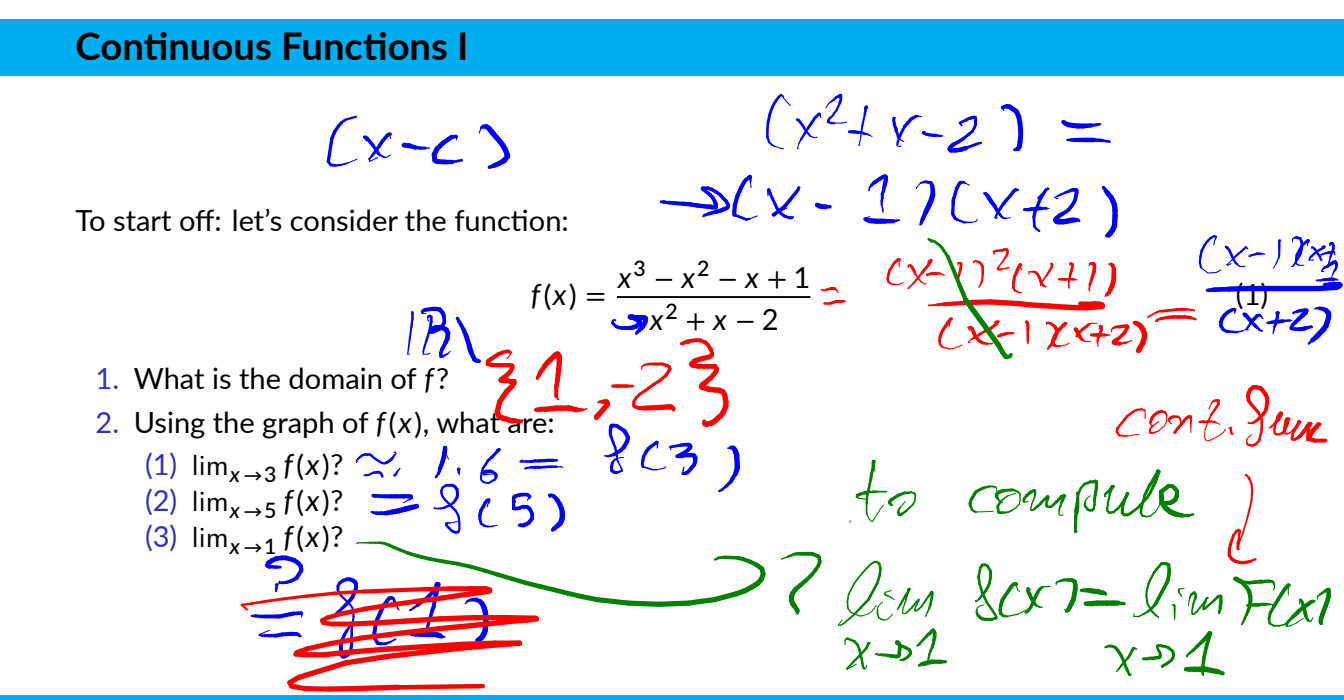
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2. Using the graph of $f(x)$, what are:
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   (2) $\lim_{x \to 5} f(x)$?
   (3) $\lim_{x \to 1} f(x)$?

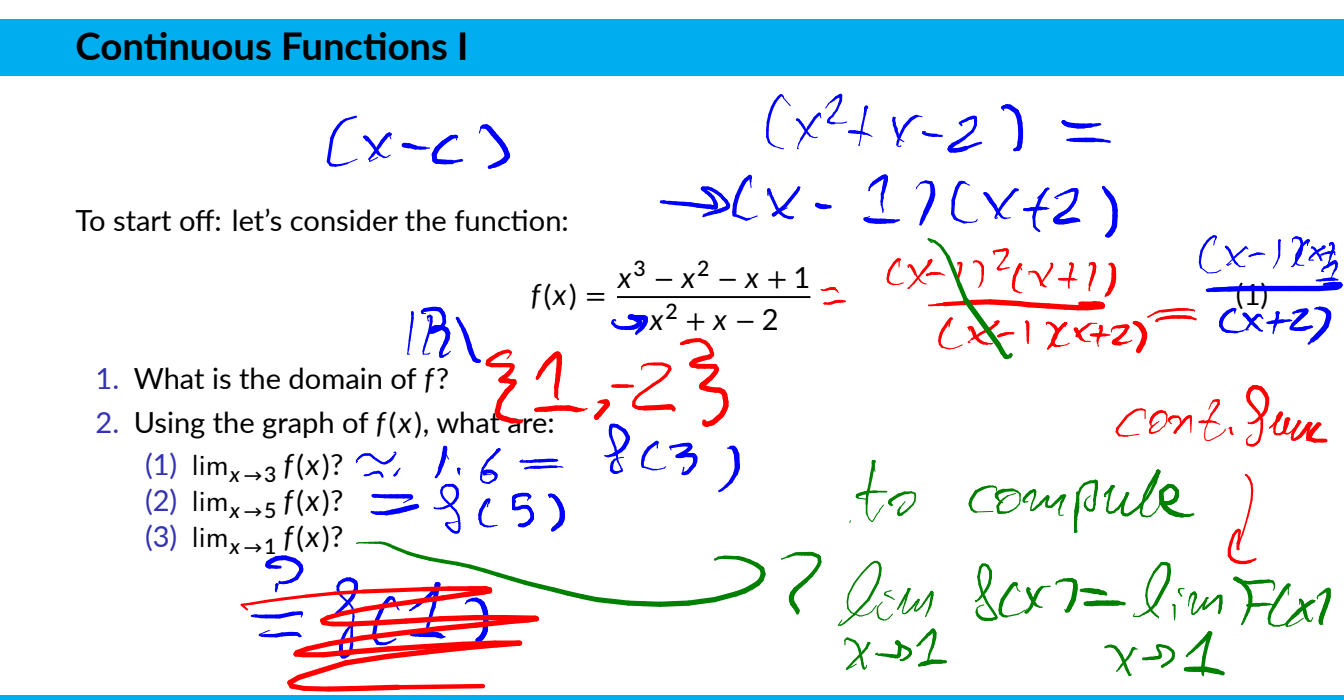


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Continuous Functions II

Those first two points, where we have

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are a special kind of point:

Continuity at a point

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3. $\lim_{x \to a} f(x) = f(a)$

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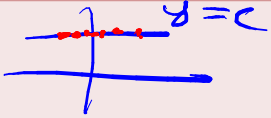
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$\lim_{x \to c} f(x) = f(c) \Rightarrow$



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4. All trigonometric functions, $\sin(x), \cos(x), \tan(x)$...

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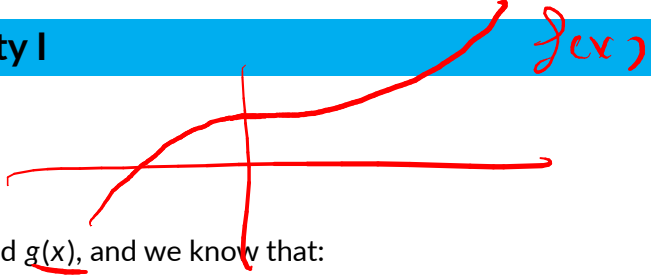
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We can determine the points at which these functions are continuous (and thus the points at which it is easy to calculate limits) by discovering *limit properties*!

Limit Properties

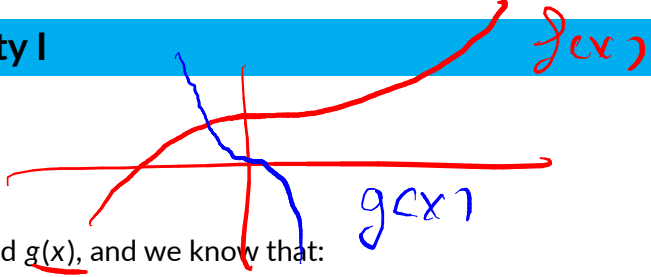
Limit Properties: Additivity I



Say we have two functions $f(x)$ and $g(x)$, and we know that:

1. As $x \to 0$, $f(x) \to 1$ and

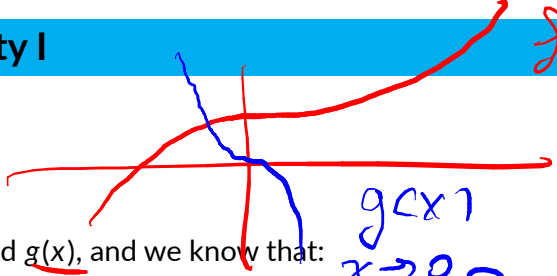
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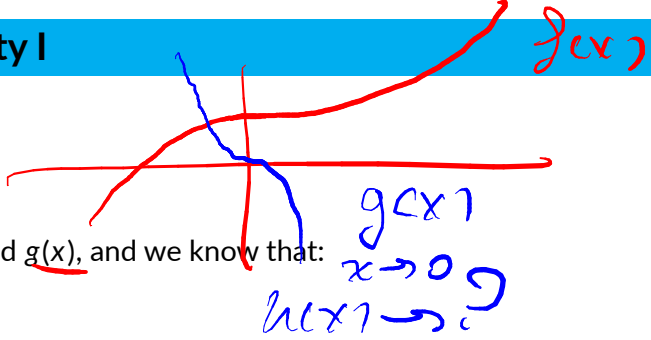
Say we have two functions $f(x)$ and $g(x)$, and we know that:

1. As $x \to 0$, $f(x) \to 1$ and
2. As $x \to 0$, $g(x) \to 0$

When $x \to 0$, towards which value does the function $h(x) = f(x) + g(x)$ approach?

1. Property 1 says that as long as $x \approx 0$ then $f(x) \approx 1$

Limit Properties: Additivity I



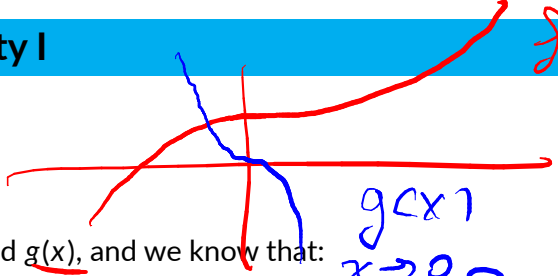
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2. Property 2 says that as long as $x \approx 0$ then $g(x) \approx 0$

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2. As $x \to 0$, $g(x) \to 0$

When $x \to 0$, towards which value does the function $h(x) = f(x) + g(x)$ approach?

1. Property 1 says that as long as $x \approx 0$ then $f(x) \approx 1$
2. Property 2 says that as long as $x \approx 0$ then $g(x) \approx 0$

So if $x \approx 0$ then $h(x) = f(x) + g(x) \approx 1 + 0 = 1$!

$$\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$$

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Let's codify this into a property:

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Then we have that $\lim_{x \to a}(f(x) + g(x))$ exists

Limit Properties: Additivity II

$g(x) = 1 + x + e^x + \sin(x)$

Let's codify this into a property:

Additivity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and a is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to b
2. $\lim_{x \to a} g(x)$ exists and is equal to c

Then we have that $\lim_{x \to a}(f(x) + g(x))$ exists and is equal to $b + c$.

Limit Properties: Multiplicativity

Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and a is a real number so that

Similarly:

Limits of Multiplicative Inverses

Limit Properties: Multiplicativity

Multiplicativity of Limits

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Then we have that $\lim_{x \to a}(f(x)g(x))$ exists and is equal to bc.

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Similarly:

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Suppose that $f(x)$ is a function and a is a real number so that $\lim_{x \to a} f(x)$ exists,

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Multiplicativity of Limits

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Then we have that $\lim_{x \to a}(f(x)g(x))$ exists and is equal to bc.

Similarly:

Limits of Multiplicative Inverses

Suppose that $f(x)$ is a function and a is a real number so that $\lim_{x \to a} f(x)$ exists, is equal to some real number b, and that $b \neq 0$.

Then we have that $\lim_{x \to a} \dfrac{1}{f(x)}$ exists

Limit Properties: Multiplicativity

Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and a is a real number so that

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Suppose that $f(x)$ is a function and a is a real number so that $\lim_{x \to a} f(x)$ exists, is equal to some real number b, and that $b \neq 0$.

Then we have that $\lim_{x \to a} \frac{1}{f(x)}$ exists and is equal to $\frac{1}{b}$.

Limit Properties: Composites

$\sin(x) - x^2$ $\quad$ $\sin(x^2)$

Limits of Composites

Suppose that $f(x)$ and $g(x)$ are two functions, that a is a real number, and that:

1. $\lim_{x \to a} f(x)$ exists

Limit Properties: Composites

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Limit Properties: Composites

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Limits of Composites

Suppose that $f(x)$ and $g(x)$ are two functions, that a is a real number, and that:

1. $\lim_{x \to a} f(x)$ exists and is equal to b
2. $f(x)$ is *not* constantly equal to b around $x = a$,

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3. $\lim_{x \to b} g(x)$ exists

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3. $\lim_{x \to b} g(x)$ exists and is equal to c

Limit Properties: Composites

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3. $\lim_{x \to b} g(x)$ exists and is equal to c

Then we have that $\lim_{x \to a} g(f(x)) = c$.

" $u = f(x)$ "

$\lim_{x \to a} g(f(x)) = \lim_{u \to b} g(u) = c$

Combining Continuous Functions

We can use the above limit laws to enlarge the class of functions we know to be continuous!

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Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number a.

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Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number a. Then we have that:

Combining Continuous Functions

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Combining Continuous Functions

Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number a. Then we have that:

1. $f(x) + g(x)$ is continuous at $x = a$

Combining Continuous Functions

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Combining Continuous Functions

Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number a. Then we have that:

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2. $f(x)g(x)$ is continuous at $x = a$
3. If $g(a)$ and $\lim_{x \to a} g(x)$ are both nonzero then $\dfrac{f(x)}{g(x)}$ is continuous at $x = a$.

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Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number a. Then we have that:

1. $f(x) + g(x)$ is continuous at $x = a$
2. $f(x)g(x)$ is continuous at $x = a$
3. If $g(a)$ and $\lim_{x \to a} g(x)$ are both nonzero then $\dfrac{f(x)}{g(x)}$ is continuous at $x = a$.
4. If $g(x)$ is instead continuous at $f(a)$ then $g(f(x))$ is continuous at $x = a$.

Moral of the Story

The big upshot of this is the following:

Strategy for Computing Limits

If you are tasked with computing a limit $\lim_{x \to a} F(x)$ algebraically

Moral of the Story



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If you are tasked with computing a limit $\lim_{x \to a} F(x)$ algebraically your goal should be to reduce the expression of the functions down

Moral of the Story



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Strategy for Computing Limits

If you are tasked with computing a limit $\lim_{x \to a} F(x)$ algebraically your goal should be to reduce the expression of the functions down to end up with an equivalent limit $\lim_{x \to a} F(x) = \lim_{x \to a} f(x)$ where:

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Strategy for Computing Limits

If you are tasked with computing a limit $\lim_{x \to a} F(x)$ algebraically your goal should be to reduce the expression of the functions down to end up with an equivalent limit $\lim_{x \to a} F(x) = \lim_{x \to a} f(x)$ where:

1. $f(x)$ is a continuous function

Moral of the Story



The big upshot of this is the following:

Strategy for Computing Limits

If you are tasked with computing a limit $\lim_{x \to a} F(x)$ algebraically your goal should be to reduce the expression of the functions down to end up with an equivalent limit $\lim_{x \to a} F(x) = \lim_{x \to a} f(x)$ where:

1. $f(x)$ is a continuous function
2. and a is in the domain of $f(x)$.

$$\lim_{x \to a} F(x) = \lim_{x \to a} g(x) = g(a)$$

Moral of the Story: Example

For instance:

is point in domain?

$$\lim_{x \to 1} \frac{x^3 - x^2 - x + 1}{x^2 + x - 2} =$$

1 ∉ domain

Moral of the Story: Example

For instance:

is point in domain?

$$\lim_{x \to 1} \frac{x^3 - x^2 - x + 1}{x^2 + x - 2} = \lim_{x \to 1} \frac{x^3 - x^2 - x - 1}{(x - 1)(x + 2)}$$

IA domain

$(x-1)$

$(x-1)$

$x^3 - x^2 - x - 1$

$x(x^2-1) - (x^2-1)$

$= (x-1)(x^2-1)$

Moral of the Story: Example

For instance:

$$\lim_{x\to 1}\frac{x^3-x^2-x+1}{x^2+x-2}=\lim_{x\to 1}\frac{x^3-x^2-x-1}{(x-1)(x+2)} \tag{2}$$

$$=\lim_{x\to 1}\frac{(x-1)(x^2-1)}{(x-1)(x+2)}$$

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$$\lim_{x\to 1}\frac{x^3-x^2-x+1}{x^2+x-2} = \lim_{x\to 1}\frac{x^3-x^2-x+1}{(x-1)(x+2)} \tag{2}$$

$$= \lim_{x\to 1}\frac{(x-1)(x^2-1)}{(x-1)(x+2)} \tag{3}$$

$$= \lim_{x\to 1}\frac{(x-1)}{(x-1)}\frac{x^2-1}{x+2}$$

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$$= \lim_{x\to 1}\frac{x^2-1}{x+2} = \tag{5}$$

$$= \frac{(1)^2-1}{1+2}$$

1& domain

1.) Is 1 in dom? ✓

2.) Is cont. @ 1?

Moral of the Story: Example

For instance:

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$$= \frac{(1)^2-1}{1+2} = \frac{1-1}{3}$$

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$$= \lim_{x\to 1}\frac{x^2-1}{x+2} = \tag{5}$$

$$= \frac{(1)^2-1}{1+2} = \frac{1-1}{3} = \frac{0}{3} = 0 \tag{6}$$

limits... but on one side

One Sided Limits

One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

One-Sided Limits

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We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

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Given a function $f(x)$ and a number a:

One Sided Limits



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Given a function $f(x)$ and a number a:

- We say that $\lim_{x \to a^-} f(x) = L$ (verbally, "the limit of $f(x)$ as x approaches *a from the left*")

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- Similarly, we say that $\lim_{x \to a^+} f(x) = L$ (verbally, "the limit of $f(x)$ as x approaches *a from the right*")

One Sided Limits



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- Similarly, we say that $\lim_{x \to a^+} f(x) = L$ (verbally, "the limit of $f(x)$ as x approaches a *from the right*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$)

One Sided Limits



We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

One-Sided Limits

Given a function $f(x)$ and a number a:

- We say that $\lim_{x \to a^-} f(x) = L$ (verbally, "the limit of $f(x)$ as x approaches a *from the left*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$) and $x < a$.
- Similarly, we say that $\lim_{x \to a^+} f(x) = L$ (verbally, "the limit of $f(x)$ as x approaches a *from the right*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$)

One Sided Limits



We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

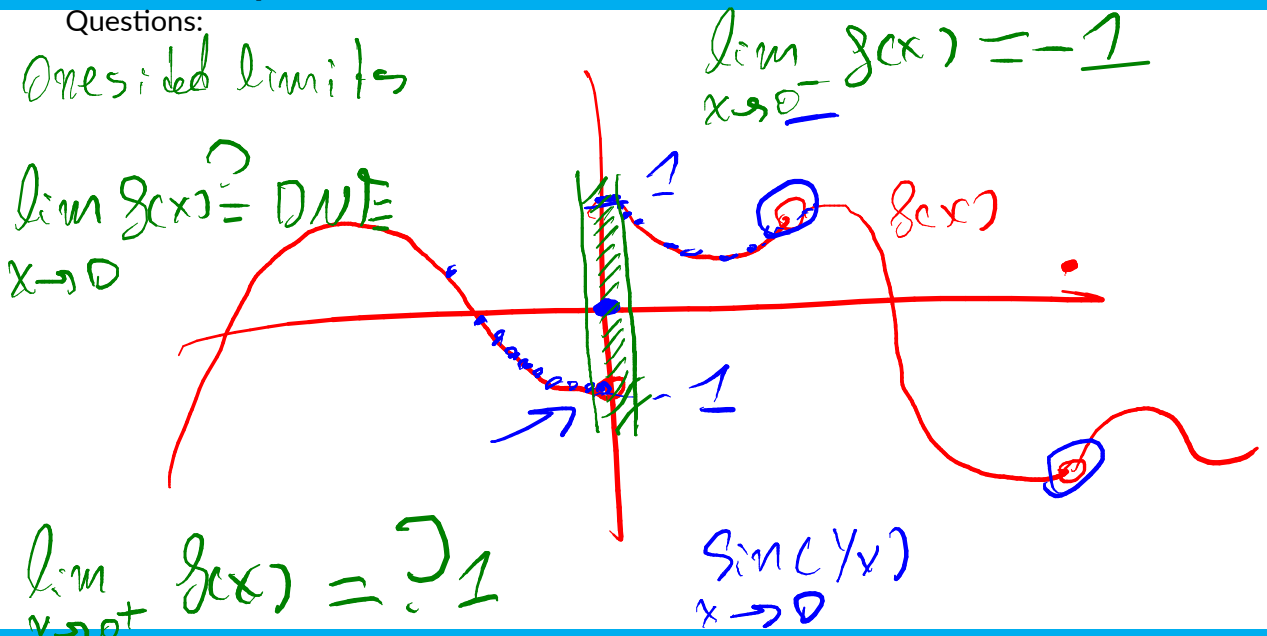
One-Sided Limits

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Relationship between Limits and One-Sided Limits

Questions:



Relationship between Limits and One-Sided Limits

Questions:

1. Can we have that $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$?

Relationship between Limits and One-Sided Limits

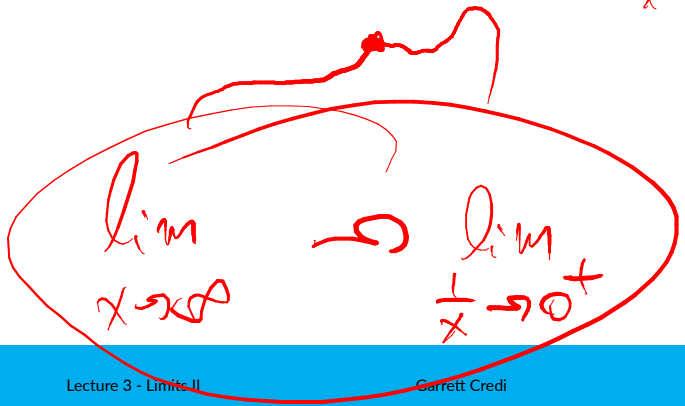
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1. Can we have that $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$?
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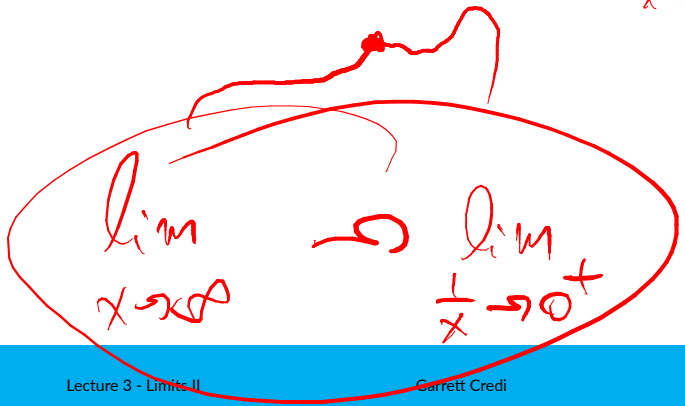
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Relationship between Limits and One-Sided Limits

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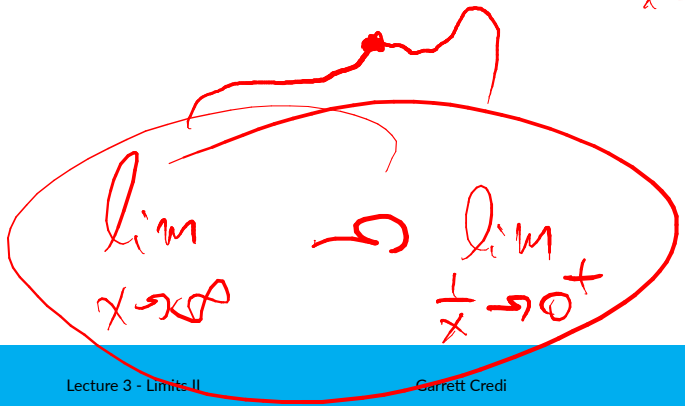
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Relationship between Limits and One-Sided Limits

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Worksheet Time!

Let’s get into groups and work on the worksheet!

End of Class!

That’s all folks!